

Deep Learning Exam

A17471E2

Hilary Term 2026

Students have 48 hours to complete this take-home exam. The solution should be submitted in the form of a report (with a maximum length of 5 pages). Each question should be answered with a detailed explanation of the mathematical and computational approach as well as a presentation of the numerical results. Code should be submitted as a separate file. The code should be written in PyTorch and use a GPU for training.

1. Consider the Heston model

$$\begin{aligned}dS_t &= rS_t dt + \sqrt{Y_t} S_t dW_t^1, \\dY_t &= \kappa(\mu - Y_t) + \sigma\sqrt{Y_t} dW_t^2,\end{aligned}\tag{1}$$

where W_t^1 and W_t^2 are Brownian motions with correlation ρ . The price of a European call option with strike price K and maturity T is

$$P(x) = E[e^{-rT} \max(S_T - K, 0)],\tag{2}$$

where $x = (S_0, Y_0, \kappa, \mu, \sigma, r, \rho, K, T)$. The price is therefore a high-dimensional function of the set of initial conditions, contract maturity and strike price, and the Heston model parameters. The objective is to use a multi-layer neural network to learn an accurate approximation to $P(x)$. By training a neural network, we can instantly evaluate/predict the option price (instead of simulating the SDE).

- (a) (10 points) Generate a training and test dataset via Monte Carlo simulation of the Heston SDE system (1). Provide a detailed description and justification for the choice of the domain for x , the numerical scheme used to simulate the SDEs, the number of Monte Carlo samples to calculate each $P(x)$, and the size of the training/test datasets.
- (b) (40 points) Train a multi-layer network $f(x; \theta)$ on the training dataset that was generated above. Provide a detailed description of the architecture and training algorithm. Evaluate its training and test

accuracy in terms of multiple metrics, including the mean absolute error, relative error, and distribution of the error. Analyze whether certain types of contracts or regions of the parameter space are difficult to approximate. Can accuracy be improved by training a larger network?

2. We now have a neural network approximator for the pricing function $P(x)$. Given observed price data, we will use the neural network approximator to calibrate the SDE parameters.

- (a) (10 points) Select specific values $\kappa^*, \mu^*, \sigma^*, r^*, \rho^*$ for the SDE parameters $\kappa, \mu, \sigma, r, \rho$ as well as specific values S_0^*, Y_0^* for the initial conditions S_0, Y_0 . For the specific values $(\kappa^*, \mu^*, \sigma^*, r^*, \rho^*, S_0^*, Y_0^*)$ generate via Monte Carlo simulation of the SDE (1) the call option prices $P_{n,m}$ for a variety of maturities and strike prices $T_1, \dots, T_N$ and $K_1, \dots, K_M$.
- (b) (20 points) Given the price dataset $P_{n,m}$, optimize over the neural network function $f(x; \theta)$ to calibrate the SDE parameters. For convenience, we can assume that the initial conditions of the SDE are known. That is, let $\nu = (\kappa, \mu, \sigma, r, \rho)$ and minimize the loss

$$L(\nu) = \frac{1}{N \times M} \sum_{n=1}^N \sum_{m=1}^M \left(f(S_0^*, Y_0^*, \nu, K_m, T_n) - P_{n,m} \right)^2. \quad (3)$$

Describe the numerical method used to minimize the loss and the selection of any hyperparameters for the method. A plot for the value of the objective function during training should be presented.

- (c) (20 points) How accurate are the calibrated parameters as compared to the true parameters $\nu^* = (\kappa^*, \mu^*, \sigma^*, r^*, \rho^*)$? How accurately does the calibrated model fit the training dataset $P_{n,m}$? In addition, generate a test dataset for contracts with different maturities and strike prices that are not included in the training dataset. What is the accuracy of the calibrated model on the test dataset?